PHY4605 Computational Methods in Physics
Learning Note | Week 11

Sensitivity and Uncertainty Propagation

Use controlled parameter variation to turn plausible input uncertainty into an output range that can be validated and interpreted physically.

Physical question. A spring is loaded by a known force, but its stiffness is only known within a plausible range. What range of extension should we report, and what does that range say about the reliability of the prediction?

Core learning outcomes

1. convert one uncertain input into lower, baseline, and upper one-at-a-time model evaluations;
2. report and validate the resulting output range without false precision; and
3. distinguish parameter uncertainty, numerical approximation error, and model limitation in plain language.

1. From a parameter sweep to uncertainty evidence

In Week 4, changing a model parameter helped reveal a physical trend. Week 11 uses the same computational move for a different purpose: the input is not known exactly, so a plausible input range becomes uncertainty evidence.

The familiar model is the ideal linear spring,

$$F = kx, \qquad x = \frac{F}{k}.$$

In words: for a fixed applied force, the predicted extension is the force divided by the spring stiffness. We use

$$F = 10\,\mathrm{N}, \qquad k_0 = 100\,\mathrm{N\,m^{-1}},$$

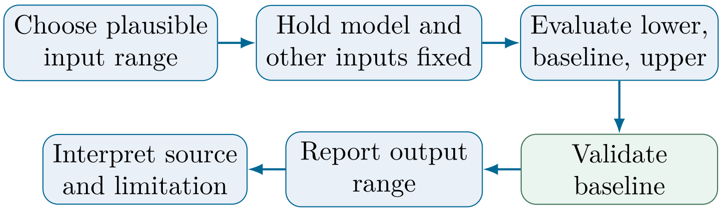
with a plausible stiffness range from 95 to $105\,\mathrm{N\,m^{-1}}$.

Quantity	Week 11 value	Meaning
F	$10\,\mathrm{N}$	applied force held fixed in the Core sweep
k	$95, 100, 105\,\mathrm{N\,m^{-1}}$	plausible lower, baseline, and upper stiffness values
x	m	model output: predicted extension

Prediction checkpoint. Because $x = F/k$, lower stiffness should produce larger extension. The $95\,\mathrm{N\,m^{-1}}$ case should therefore give the largest x, while the $105\,\mathrm{N\,m^{-1}}$ case should give the smallest.

2. One-at-a-time uncertainty reasoning

A one-at-a-time sweep changes one uncertain input while keeping the model and every other input fixed. This allows the resulting output change to be attributed to that one input range.



The computational recipe is:

1. define the model, baseline, units, and one plausible input range;
2. hold all other inputs and assumptions fixed;
3. evaluate the lower, baseline, and upper cases;
4. collect the minimum and maximum output;
5. validate one known or hand-calculated case; and
6. state what the output range means physically.

Do not vary several uncertain inputs at once when the question asks which input caused the change. Simultaneous variation can be useful for other analyses, but it no longer isolates one input's effect.

3. MATLAB scaffold: three controlled cases

The Core calculation needs only a short array and element-wise division. The three stiffness values form three controlled evaluations of the same deterministic equation.

```
force_baseline_N = 10;
stiffness_baseline_Npm = 100;
stiffness_cases_Npm = [95 100 105];

extension_cases_m = force_baseline_N./stiffness_cases_Npm;
extension_baseline_m = force_baseline_N/stiffness_baseline_Npm;
extension_range_m = [min(extension_cases_m) max(extension_cases_m)];
```

The dot in `./` means that the scalar force is divided by each stored stiffness value. The calculation gives

$$x_0 = 0.10000\,\mathrm{m},$$

and the range

$$0.09524\,\mathrm{m} \lesssim x \lesssim 0.10526\,\mathrm{m}.$$

4. Report a range, not just a long decimal

The exact-looking statement "$x = 0.10000\,\mathrm{m}$" describes only the baseline case. It hides the fact that the stiffness itself is uncertain over the supplied range.

A more defensible statement is:

Range-based result. For a 10 N load and stiffness between 95 and 105 $\mathrm{N\,m^{-1}}$, the ideal Hooke model predicts an extension of about 0.095 to 0.105 m.

Extra displayed decimal places do not improve knowledge of the spring stiffness. Numerical display precision and physical parameter uncertainty are different ideas.

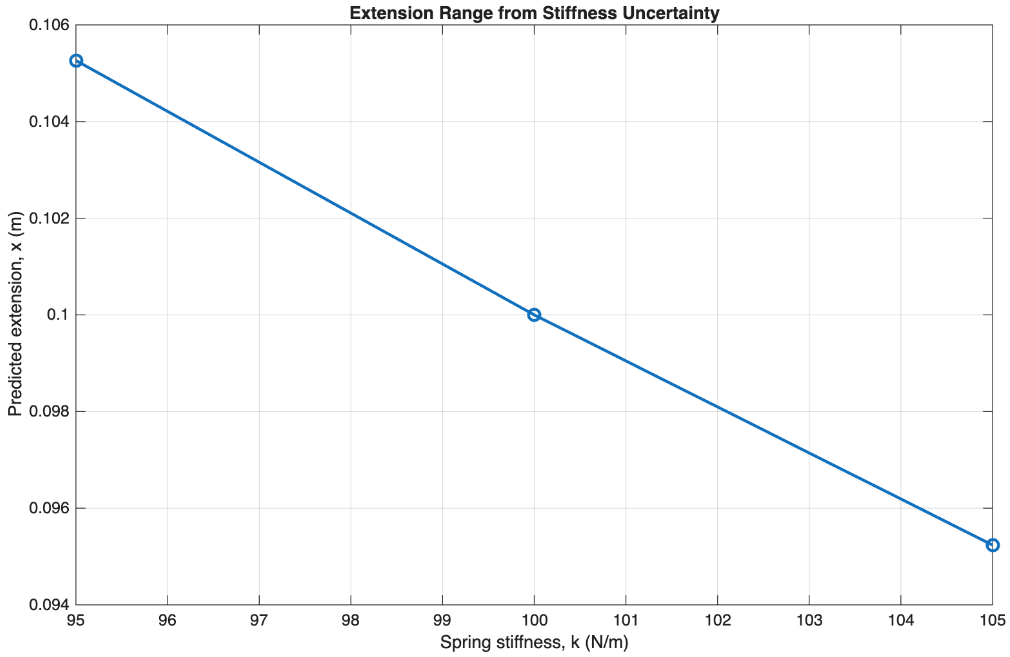


Figure 1: Predicted extension for the supplied stiffness values. These points are controlled parameter cases, not repeated random trials.

5. Core validation: check the baseline independently

The baseline is simple enough to calculate by hand:

$$x_0 = \frac{10\,\mathrm{N}}{100\,\mathrm{N\,m^{-1}}} = 0.10\,\mathrm{m}.$$

The units also reduce correctly:

$$\frac{\mathrm{N}}{\mathrm{N\,m^{-1}}} = \mathrm{m}.$$

```
reference_baseline_m = 10/100;
baseline_check_passed = abs(extension_baseline_m-reference_baseline_m) < 1e-12;
assert(baseline_check_passed,'Baseline Hooke-law validation failed.')
```

Validation result. The baseline matches the independent calculation, the units are metres, and the trend is physically consistent: decreasing k increases x at fixed F. A smooth graph alone would not establish these checks.

6. Three different reasons a prediction can be imperfect

The word "error" is too vague unless its source is identified.

Source	Meaning	Week 11 example
Parameter uncertainty	an input is not known exactly	spring stiffness is plausible from 95 to 105 $\mathrm{N\,m^{-1}}$
Numerical approximation error	a mathematical operation is approximated numerically	a finite timestep changes an ODE simulation result
Model limitation	the equation's assumptions do not fully represent reality	Hooke's law may fail outside the linear elastic range

The direct Core Hooke calculation has no meaningful timestep, grid, or iterative approximation. Its uncertainty range comes from the stated stiffness range. Even a perfectly evaluated equation can still inherit uncertainty from its inputs and limitations from its assumptions.

7. Worked reasoning chain

Step	Week 11 evidence
Physical model	ideal linear spring, $x = F/k$
Scale and units	F in N, k in N/m, x in m
Uncertain input	$k = 95, 100, 105\,\mathrm{N\,m^{-1}}$
Algorithm	hold F fixed, evaluate the same model for lower/baseline/upper k
Output range	approximately 0.09524 to 0.10526 m
Validation	hand-check $10/100 = 0.10\,\mathrm{m}$ plus units and trend
Interpretation	stiffness uncertainty creates a range of plausible model predictions

This chain is more important than memorising a MATLAB function name. In an assessment, syntax may be supplied; the physical model, held-fixed conditions, uncertainty source, validation, and interpretation still need to be defended.

Working exposure: compare a second uncertain input

Now hold $k = 100\,\mathrm{N\,m^{-1}}$ fixed and vary the force from 9.8 to 10.2 N. The corresponding extension range is 0.098 to 0.102 m, with a full span of 0.004 m.

For the stiffness range, the full extension span is about 0.01003 m. For the supplied uncertainty ranges, stiffness therefore changes the output more than force. This conclusion depends on the ranges that were supplied; it is not a universal statement that stiffness must always dominate force.

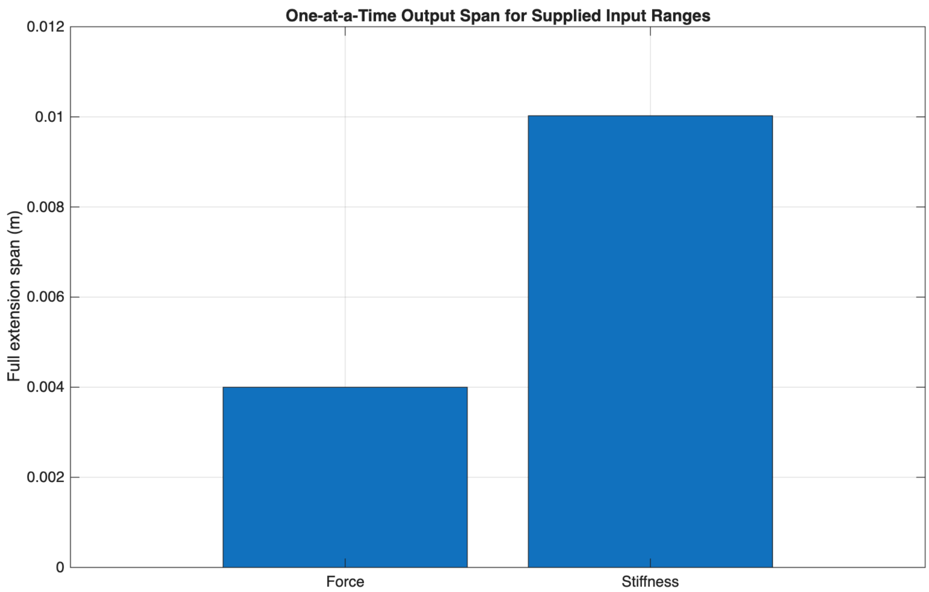


Figure 2: One-at-a-time output spans for the supplied stiffness and force ranges. The comparison isolates one input range at a time.

Working exposure: supplied sensitivity and propagation formulas

A supplied normalized sensitivity can compare fractional output change with fractional input change near a baseline:

$$S \approx \frac{\Delta x/x_0}{\Delta p/p_0}.$$

For $x = F/k$, the supplied local interpretation is approximately $S_k = -1$ and $S_F = +1$. The sign gives the direction of response; the magnitude describes the fractional responsiveness near the baseline. Deriving these sensitivities is not Core Week 11 work.

If small force and stiffness uncertainties are treated as independent standard uncertainties, a supplied first-order relation is

$$\frac{u_x}{x} \approx \sqrt{\left(\frac{u_F}{F}\right)^2 + \left(\frac{u_k}{k}\right)^2}.$$

Using $u_F = 0.2\,\mathrm{N}$ and $u_k = 5\,\mathrm{N\,m^{-1}}$ gives a relative estimate of about 5.39% and

$$u_x \approx 0.00539\,\mathrm{m}.$$

This combined estimate answers a different question from the isolated one-at-a-time ranges. The formula is supplied for interpretation only.

Optional stretch

Independent derivative-based propagation, Monte Carlo uncertainty propagation, correlated inputs, interaction effects, and global sensitivity methods are outside the Week 11 Core route.

Three takeaways

1. An uncertain input should be represented by a justified baseline and plausible range before evaluating the model.
2. A one-at-a-time sweep converts that input range into an interpretable output range while keeping other inputs fixed.
3. Parameter uncertainty, numerical approximation error, and model limitation describe different reasons a prediction may be imperfect.

Preparation for the practical. Be ready to identify one uncertain input and its units, predict the direction of the output change, compute a lower/baseline/upper range using a supplied model, perform one independent check, and explain the result without false precision.